



yes add date time log for better retention and in the folders change the file name like in auth folder we do need to write auth.controller.ts we can directly write controller.ts

Perfect, both changes fit nicely into the design.

Date-time in logs + file retention

We will add **timestamp** to every log entry and also include date in log file names for better retention and auditing.

1. Per-log timestamp

In Winston, we'll configure a global format like:

```
import { createLogger, format, transports } from 'winston';

const logger = createLogger({
  level: 'info',
  format: format.combine(
    format.timestamp({ format: 'YYYY-MM-DD HH:mm:ss' }), // or Asia/Kolkata via a helper
    format.json()
  ),
  transports: [
    new transports.File({ filename: 'logs/app.log' }),
    new transports.Console(),
  ],
});
```

`format.timestamp` adds a `timestamp` field to every log entry, and we can format it as `YYYY-MM-DD HH:mm:ss` for easy reading. [\[1\]](#) [\[2\]](#) [\[3\]](#) [\[4\]](#)

If you prefer, we can customize to IST explicitly using a small helper function that formats timestamps in Asia/Kolkata.

2. Date-based log files

For retention, we'll use a rotating file transport (e.g. `winston-daily-rotate-file`) to generate files like:

- `logs/app-2026-06-18.log`
- `logs/auth-2026-06-18.log`

This makes it very easy to keep daily archives and comply with “legal issue” audits later. [\[3\]](#) [\[5\]](#)

Feature-based folder naming (short filenames)

We'll follow your preference: inside each feature folder, use generic filenames like `controller.ts`, `service.ts`, etc., and rely on the folder name (`auth`, `buses`, `bookings...`) to indicate the domain. [\[6\]](#) [\[7\]](#) [\[8\]](#)

Updated structure:

```
src/
  config/
    env.ts
    db.ts
    redis.ts
    logger.ts
  core/
    middleware/
      auth.middleware.ts
      role.middleware.ts
      error.middleware.ts
      cors.middleware.ts
      logging.middleware.ts
      validate.middleware.ts
    utils/
      jwt.ts
      response.ts
  features/
    auth/
      controller.ts
      service.ts
      routes.ts
      validators.ts
      types.ts
    users/
      controller.ts
      service.ts
      routes.ts
      validators.ts
    buses/
      controller.ts
      service.ts
      routes.ts
      validators.ts
  routes/
```

```

    controller.ts
    service.ts
    routes.ts
    validators.ts
  schedules/
    controller.ts
    service.ts
    routes.ts
    validators.ts
  seats/
    controller.ts
    service.ts
    routes.ts
    validators.ts
  search/
    controller.ts
    service.ts
    routes.ts
    validators.ts
  bookings/
    controller.ts
    service.ts
    routes.ts
    validators.ts
  payments/
    controller.ts
    service.ts
    routes.ts
    validators.ts
  logs/
    controller.ts      # list logs, stream logs
    routes.ts
  prisma/
    schema.prisma
  app.ts
  server.ts

```

This keeps paths very readable:

- features/auth/controller.ts
- features/bookings/service.ts

and still respects clear separation of concerns. The combination of **feature folder name + file type** is in line with modern Express/Node structuring recommendations.^{[7] [8] [9]}

If you're okay with this, next we can:

1. lock this structure as “final”, and
2. move to designing the Prisma schema (Step 1) with these conventions in mind.

✱✱

1. <https://stackoverflow.com/questions/10271373/node-js-how-to-add-timestamp-to-logs-using-winston-library>.
2. <https://stackoverflow.com/questions/13987631/winstonhow-to-change-timestamp-format/51553641>

3. <https://gist.github.com/Avi-E-Koenig/5f4578bb3fea272aaba67ba5fb53f871>
4. <https://fatehaliaamir.medium.com/winston-the-ultimate-logger-91c911171f5d>
5. <https://github.com/winstonjs/winston/issues/1663>
6. https://dev.to/pramod_boda/recommended-folder-structure-for-nodets-2025-39jl
7. <https://dev.to/ehtisamhaq/exploring-design-patterns-for-expressjs-projects-mvc-modular-and-more-37lf>
8. <https://albertobasalo.medium.com/file-and-folder-structure-for-node-express-applications-d7d5ed32b8c5>
9. <https://copyprogramming.com/howto/winston-how-to-change-timestamp-format>
10. <https://github.com/ZhongkaiChen/express-controllers>
11. <https://stackoverflow.com/questions/56826988/winston-how-to-get-logging-with-timestamp-at-the-front>
12. <https://stackoverflow.com/questions/64792757/winston-logger-custom-format-overwriting-timestamp>
13. https://www.linkedin.com/posts/sahinur_nodejs-expressjs-webdevelopment-activity-7344062772667695104-CUz0
14. <https://stackoverflow.com/questions/58956937/express-js-in-which-folder-does-a-controller-js-belong>
15. <https://medium.com/codechef-vit/a-better-project-structure-with-express-and-node-js-c23abc2d736f>